

Juan C. Yamin

Department of Economics, Brown University

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2026–27 Economics Job Market Candidate

Fields: Econometrics, Development Economics
Research interests: Statistical decision theory, experimental design, causal inference
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DISSERTATION COMMITTEE AND REFERENCES

Toru Kitagawa (Chair)
Professor of Economics
Brown University
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Soonwoo Kwon
Assistant Professor of Economics
Brown University
soonwoo_kwon@brown.edu

Jonathan Roth
Professor of Economics
Brown University
jonathan_roth@brown.edu

EDUCATION

Brown University, Providence, RI

Ph.D. in Economics

M.A. in Economics

Expected May 2027

May 2022

Universidad de los Andes, Bogotá, Colombia

M.Sc. in Economics

B.Sc. in Economics

March 2019

October 2017

JOB MARKET PAPER

Poverty Targeting with Imperfect Information

I study how antipoverty programs should allocate transfers when income is known only through noisy predictions. I show that the standard plug-in rule is inadmissible. I then develop a nonparametric empirical Bayes rule whose Bayes regret is bounded by the mean squared error in estimating oracle posterior mean poverty gaps. In simulations using surveys from nine African countries, the rule reaches about 1.8 times as many poor people as plug-in OLS with the same budget and achieves the same average poverty-gap reduction with 6.7 percent less spending.

OTHER WORKING PAPERS

When and How to Pilot: Design Rules for Two-Wave Experiments

I propose a Conditional Minimax Regret (CMR) rule for treatment assignment in two-wave experiments. CMR minimizes worst-case regret over a finite-sample confidence set for treatment and control variances. It retains the worst-case protection of balanced assignment with high probability, converges to the Neyman allocation as the pilot grows, and attains the minimax-regret rate up to constants. In simulations calibrated to four field experiments, it avoids the severe small-pilot losses of feasible Neyman allocation while matching its large-pilot gains.

Two-Way Effects Models: A Nonparametric Empirical Bayes Approach

with Cole Davis

Draft available upon request

We develop nonparametric empirical Bayes shrinkage estimators for AKM worker and firm effects. We also estimate how the mean and dispersion of worker effects vary with firm wage premia, accounting for correlated estimation errors.

PUBLICATIONS

Birds of a Feather Collude Together: Subnational Alignment and Corruption

with Leopoldo Fergusson, Arturo Harker, and Carlos Molina

Forthcoming in the American Political Science Review

Using a regression discontinuity design in Colombia, we find that partisan alignment between mayors and governors increases the share of “ghost” students by about 0.3 standard deviations, inflating education transfers without improving genuine enrollment or student performance. Alignment also increases patronage and electoral fraud risk, while aligned politicians enjoy better future electoral prospects.

SOFTWARE

cmrdesign (R and Python)

[Documentation](#) | [Code](#) | [CRAN](#) | [PyPI](#)

Uses pilot data to choose main-wave treatment and control sample shares and report a finite-sample regret certificate. Supports two-arm, multi-arm, and stratified experiments.

RESEARCH AND INDUSTRY EXPERIENCE

Brown University, Department of Economics

June 2023–Jan. 2026

Research Assistant to Soonwoo Kwon, Providence, RI

Conducted simulations and empirical applications on fixed-effect shrinkage, regression discontinuity inference under monotonicity, and bias-aware inference in regularized regression.

Amazon, Delivery Experience

May–Aug. 2025

Economist Intern, Bellevue, WA

Led a research project evaluating a nationwide U.S. delivery-speed program. Acquired data using SQL and implemented difference-in-differences and shift-share IV designs in Python to estimate direct and geographic spillover effects on customer delivery experience. Presented findings to senior economists and program decision-makers.

Universidad de los Andes, Department of Economics

Aug. 2017–July 2021

Research Assistant to Leopoldo Fergusson and James A. Robinson, Bogotá, Colombia

Conducted empirical analysis for Acemoglu, Fergusson, and Johnson's study of population growth and conflict, published in the *Review of Economic Studies*. Built and linked administrative, archival, and web-scraped datasets across projects on conflict, corruption, and state capacity. Applied instrumental-variables and regression-discontinuity designs and estimated panel-data models with two-way fixed effects.

CONFERENCE PRESENTATIONS

Conference on Digital Experimentation (CODE@MIT), Cambridge

2026

When and How to Pilot: Design Rules for Two-Wave Experiments (accepted for presentation)

Midwest Econometrics Group Annual Meeting, Cincinnati

2026

Poverty Targeting with Imperfect Information (accepted for presentation)

World Congress of the Econometric Society, Seoul

2025

Poverty Targeting with Imperfect Information

Advances with Field Experiments, Chicago

2025

When and How to Pilot: Design Rules for Two-Wave Experiments

AWARDS AND FELLOWSHIPS

Merit Dissertation Fellowship, Department of Economics, Brown University

Fall 2026

International Travel Fund, Graduate School, Brown University

2025

Conference Travel Fund, Graduate School, Brown University

2025

TEACHING EXPERIENCE

Brown University, Teaching Assistant

Applied Econometrics II (Ph.D.), Peter Hull

Spring 2026

Applied Econometrics I (Ph.D.), Toru Kitagawa

Fall 2024

Using Big Data to Solve Economic and Social Problems (undergraduate), John N. Friedman

Fall 2022 and 2023

Universidad de los Andes, Teaching Assistant

Aug. 2018–June 2020

Advanced Econometrics (graduate), Raquel Bernal

Political Underpinnings of Prosperity and Poverty (graduate), James A. Robinson

Thinking Problems (undergraduate), Tomás Rodríguez

ACADEMIC SERVICE

Organizer, Brown Metrics Coffee

2024–26

TECHNICAL SKILLS AND LANGUAGES

Tools: R, Python, Stata, MATLAB, SQL, Git, LaTeX

Languages: Spanish (native), English (fluent), French (fluent)